

Law of large numbers

Imagine flipping a coin repeatedly. Let X_i be the indicator variable for getting a head on the i th flip. Then $X_i \sim \text{Bernoulli}\left(\frac{1}{2}\right)$. Assuming independent flips, we have a sequence $X_1, X_2, X_3, \dots$ of independent and identically distributed (iid) random variables.

1. The random variables X_i all have the same distribution. What are $E[X_i]$ and $\text{Var}(X_i)$?

$$E[X_i] = \frac{1}{2}$$

$$\text{Var}(X) = \frac{1}{4}$$

2. We write $\bar{X}_n$ for $(X_1 + \dots + X_n)/n$, which is a random variable. For any outcomes of the repeated coin flips, the output of this random variable is the proportion of heads in the first n flips. Calculate $E[\bar{X}_{10}]$, $\text{Var}(\bar{X}_{10})$, and $\sigma(\bar{X}_{10})$.

$$E[\bar{X}_{10}] = \frac{1}{2}$$

$$\text{Var}(\bar{X}_{10}) = \frac{1}{40}$$

$$\sigma(\bar{X}_{10}) = \frac{1}{\sqrt{40}}$$

3. Let n be an arbitrary positive integer. Calculate $E[\bar{X}_n]$, $\text{Var}(\bar{X}_n)$, and $\sigma(\bar{X}_n)$.

$$E[\bar{X}_n] = \frac{1}{2}$$

$$\text{Var}(\bar{X}_n) = \frac{1}{4n}$$

$$\sigma(\bar{X}_n) = \frac{1}{\sqrt{4n}}$$

4. Describe what happens to these three quantities as $n \rightarrow \infty$.

5. Use the two Law of Large Numbers tools on Blackboard to answer the following questions.

How many flips does it seem to take to get proportion of heads typically within 0.02 of 0.5, i.e. between 0.48 and 0.52?

How many flips does it take for the probability of the proportion of heads within 0.02 of 0.5 at least 0.9? At least 0.95? At least 0.99?

Answer these questions again but making the proportion of heads within 0.01 of 0.5.